MR. ALLEN'S 8TH GRADE MATH

Q1 Benchmark Review

Quarter 1

Name _____ Date _____ Period _____

PART 1 The Real Number System

NC.8.NS.1 · NC.8.NS.2

REMEMBER **Rational** = can be written as a fraction of integers (decimals that end or repeat, √ of perfect squares). **Irrational** = decimal never ends and never repeats ($\sqrt{2}$, π). Rational + irrational → **irrational**. Rational (not 0) × irrational → **irrational**.

1 Select **all** the numbers that are **irrational**. SELECT ALL

Ⓐ $\sqrt{64}$ Ⓑ $\sqrt{18}$ Ⓒ $0.\overline{6}$

Ⓓ 3.14159 Ⓔ $\frac{\pi}{2}$ Ⓕ $-\sqrt{25}$

2 Is the value of each expression rational or irrational? Check one box in each row. TABLE

Expression	Rational	Irrational
$6 + \sqrt{11}$	☐	☐
$\sqrt{5} \cdot \sqrt{5}$	☐	☐
4π	☐	☐
$0.25 + \frac{3}{4}$	☐	☐

3 Estimate $\sqrt{55}$ to the nearest **tenth**. Do not use a calculator. SHORT ANSWER

It is between the whole numbers _____ and _____

Perfect squares on each side: _____ and _____

ANSWER ≈ _____

4 Order these numbers from **least to greatest**. SHORT ANSWER

$\sqrt{20}$ $4.\overline{4}$ $\frac{9}{2}$ $\pi + 1$ 4.38

Least _____ _____ _____ _____ _____ Greatest

PART 2 Exponent Rules with Monomials

NC.8.EE.1

REMEMBER Numbers with numbers, letters with letters. **Multiply → ADD** exponents. **Divide → SUBTRACT** exponents. **Power to a power → MULTIPLY** exponents, and raise the number to the power too.

5 Simplify. $(-3x^4y^2)(5x^3y)$ SHORT ANSWER

ANSWER

6 Simplify. $\frac{24a^8b^3}{6a^2b^3}$ SHORT ANSWER

ANSWER

7 Which expression is equivalent to $(-2x^3)^4$? MULTIPLE CHOICE

Ⓐ $-8x^7$ Ⓑ $-16x^{12}$ Ⓒ $16x^{12}$ Ⓓ $8x^{12}$

8 Simplify. $(2m^3)^2 \cdot 3m^4$ SHORT ANSWER

ANSWER

Turn over for Part 3 →

Calculator inactive: Parts 1 and 2

PART 3 Scientific Notation

NC.8.EE.3 · NC.8.EE.4

REMEMBER The first number must be **at least 1 and less than 10**. Multiply → multiply the numbers, **ADD** the exponents. Divide → divide the numbers, **SUBTRACT** the exponents. Then check that the first number is still from 1 to less than 10.

9 Write each number in scientific notation. SHORT ANSWER

a. **3,040,000,000** = _____ × 10 ___

b. **0.000062** = _____ × 10 ___

10 Each number is **not** in correct scientific notation. Rewrite it correctly. SHORT ANSWER

a. 48.5×10^{6} = _____ × 10 ___

b. 0.73×10^{-3} = _____ × 10 ___

11 Multiply. Write your answer in scientific notation. SHORT ANSWER

$$(4 \times 10^{5})(6 \times 10^{3})$$

ANSWER _____ × 10 ___

12 Divide. Write your answer in scientific notation. SHORT ANSWER

$$\frac{3.6 \times 10^{4}}{9 \times 10^{-2}}$$

ANSWER _____ × 10 ___

13 One grain of rice has a mass of about 2.5×10^{-2} grams. A warehouse has 4×10^{6} grains of rice. What is the total mass of the rice, in grams? Write your answer in scientific notation. WORD PROBLEM

ANSWER _____ × 10 ___ grams

14 An elephant has a mass of about 6×10^{3} kilograms. A mouse has a mass of about 2×10^{-2} kilograms. About how many times as great is the mass of the elephant as the mass of the mouse? WORD PROBLEM

ANSWER _____ × 10 ___ times

15 Which statement is **true**? MULTIPLE CHOICE

(A) $7.2 \times 10^{-4} > 3.1 \times 10^{-3}$

(B) $5 \times 10^{6} < 9 \times 10^{5}$

(C) $2.8 \times 10^{7} > 9.9 \times 10^{6}$

(D) $4.5 \times 10^{3} > 4.5 \times 10^{4}$